Tutorial Session: K-Means Clustering and PCA

What is K-Means?

K-Means is an unsupervised machine learning algorithm used for clustering data points into groups.

The algorithm divides the dataset into K clusters such that:

- Data points inside the same cluster are similar.
- Data points in different clusters are dissimilar.
- Similarity is measured using Euclidean distance.

Intuition Behind K-Means

Suppose we have customer data and want to group customers with similar behavior.

K-Means works as follows:

1. Randomly select K centroids.
2. Assign each point to the nearest centroid.
3. Update centroid positions.
4. Repeat until cluster assignments stop changing.

Step 1: Choose K

Examples:

- $K = 2$ means 2 clusters.
- $K = 3$ means 3 clusters.

Step 2: Initialize Centroids Select K random points as centroids.

Step 3: Assign Points to Nearest Centroid Euclidean distance formula:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

For 3-dimensional data:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Step 4: Update Centroids New centroid:

$$C = \left(\frac{\sum x_i}{n}, \frac{\sum y_i}{n}, \frac{\sum z_i}{n} \right)$$

Step 5: Repeat Continue assigning and updating until clusters stop changing.

Dataset for K-Means and PCA

ID	X	Y	Z
1	1	2	1
2	2	1	2
3	1	1	1
4	8	9	8
5	9	8	9
6	8	8	9
7	4	5	4
8	5	4	5
9	4	4	5
10	5	5	4

Question 1: Initial Clustering

Take:

- $K = 2$
- Initial centroids:
  - $C_1 = (1, 2, 1)$
  - $C_2 = (8, 9, 8)$

Tasks:

1. Compute Euclidean distance for each point.
2. Assign clusters.
3. Find updated centroids.

Question 2: Effect of Changing K

Using the same dataset:

1. Apply K-Means with $K = 2$.
2. Apply K-Means with $K = 3$.
3. Compare:
   - Number of clusters
   - Cluster compactness
   - Points changing clusters

Question 3: Final Cluster Identification

Suppose final centroids are:

$$C_1 = (1.33, 1.33, 1.33)$$

$$C_2 = (4.5, 4.5, 4.5)$$

$$C_3 = (8.33, 8.33, 8.67)$$

Tasks:

1. Identify cluster membership.
2. Explain why the grouping makes sense.

Question 4: Outlier Effect

Add the point:

$$(20, 20, 20)$$

Tasks:

1. Apply K-Means with $K = 3$.
2. Observe centroid movement.
3. Explain effect of outliers.

Question 5: Distance Computation

Find Euclidean distance between:

$$A = (1, 2, 1)$$

and

$$B = (5, 5, 4)$$

Solution setup:

$$d = \sqrt{(5-1)^2 + (5-2)^2 + (4-1)^2}$$

Advantages of K-Means

- Simple and fast
- Easy to implement
- Works well for spherical clusters
- Efficient for large datasets

Limitations of K-Means

- Need to choose K manually
- Sensitive to initialization
- Sensitive to outliers
- Poor performance on non-spherical clusters

Choosing Optimal K

One common method is the Elbow Method.
Idea:

- Compute Within Cluster Sum of Squares (WCSS)
- Plot WCSS vs K
- Choose elbow point

What is PCA?

PCA is a dimensionality reduction technique.
It transforms high-dimensional data into fewer dimensions while preserving maximum variance.
Example:

- Original features $= 3$
- PCA output $= 2$ principal components

Why PCA?

PCA helps in:

- Reducing computation
- Removing redundancy
- Visualization of high-dimensional data
- Noise reduction

Intuition Behind PCA

Suppose features are highly correlated.
Instead of using all features separately, PCA creates new variables called Principal Components.
These:

- Are uncorrelated
- Capture maximum variance

Step 1: Standardization

Standardization formula:

$$z = \frac{x - \mu}{\sigma}$$

where:

- μ = mean
- σ = standard deviation

Step 2: Covariance Matrix

Covariance formula:

$$Cov(X, Y) = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{n - 1}$$

Step 3: Eigenvalues and Eigenvectors

- Eigenvectors give principal directions.
- Eigenvalues represent importance of those directions.

Largest eigenvalue corresponds to Principal Component 1.

Step 4: Select Principal Components

Example:

- PC1 captures 85% variance
- PC2 captures 10% variance
- PC3 captures 5% variance

Then retain only PC1 and PC2.

Step 5: Transform Data

Project original data onto selected principal components.

Question 1: Mean Calculation

Compute:

- Mean(X)
- Mean(Y)
- Mean(Z)

Question 2: Standardization

For row 1:

$$(1, 2, 1)$$

Tasks:

1. Compute standardized values.
2. Explain importance of standardization.

Question 3: Covariance Interpretation

Suppose covariance matrix is:

	X	Y	Z
X	1	0.92	0.95
Y	0.92	1	0.90
Z	0.95	0.90	1

Tasks:

1. Identify strongly correlated features.
2. Explain why PCA is useful.

Question 4: Principal Component Selection

Suppose eigenvalues are:

Component	Eigenvalue
PC1	2.7
PC2	0.2
PC3	0.1

Tasks:

1. Which component captures maximum variance?
2. How many PCs should be retained?
3. Compute variance percentage captured by PC1.

Formula:

$$Variance\ \% = \frac{Eigenvalue}{Total\ \ Eigenvalues} \times 100$$

Question 5: Dimensionality Reduction

Original dataset:

- 3 dimensions

After PCA:

- Reduced to 2 dimensions

Tasks:

1. What information may be lost?
2. What are advantages of dimensionality reduction?

Question 6: PCA for Visualization

Explain:

1. Why PCA is useful for visualization.
2. How 100-dimensional data can be visualized in 2D.

Question 1

Can PCA be applied before K-Means?
Explain why.

Question 2

Why is feature scaling important in both K-Means and PCA?

Question 3

Suppose one feature has values from 1–10 and another from 1–10000.

Tasks:

1. Explain impact on K-Means.
2. Explain impact on PCA.
3. Why should normalization or standardization be used?

K-Means Summary

- Unsupervised clustering algorithm
- Groups similar data points
- Uses Euclidean distance
- Requires choosing K

PCA Summary

- Dimensionality reduction technique
- Finds principal components
- Preserves maximum variance
- Helps visualization and compression